

Customer Engagement & Product Utilization Analytics for Retention Strategy

A Behavioral Analysis of Banking Customer Churn: Why Engagement and Product Depth Predict Retention Better Than Balance or Salary Alone

Prepared for: Unified Mentor | The European Central Bank
Dataset: European_Bank.csv — 10,000 customer records
Research Paper — Internship Project Deliverable

1. Abstract

Banks often assume that financially strong customers — those with high balances or salaries — are inherently loyal, yet churn frequently occurs regardless of financial standing. This study analyzes 10,000 retail banking customers to test whether behavioral engagement and product utilization predict retention better than demographic or financial attributes alone. Customers were segmented into four engagement profiles, and five KPIs were developed to quantify engagement-driven retention risk. Key findings show that active, multi-product customers churn at 9.7%, compared to 27.1% among inactive customers — a 2.8x gap. Product count shows a U-shaped, not linear, relationship with churn: two products is the stickiest tier (7.6% churn), while three to four products carries near-total churn (82.7%–100%). A further 1,247 customers (12.5% of the base) hold 24.3% of total deposits while disengaged, representing concentrated financial exposure. A Random Forest model corroborates these findings while surfacing customer age as an unmodeled predictive factor. The results support engagement-first retention strategy over demographic-based targeting.

2. Background & Problem Statement

Banks increasingly recognize that customer behavior and engagement — not just demographics — determine long-term retention. Customers may appear financially strong, carrying a high balance or salary, yet still churn due to low engagement, limited product adoption, or a weak overall relationship with the bank. Understanding how customers actually use banking products and services is essential to designing effective cross-sell strategies, loyalty programs, and engagement-driven retention initiatives.

Despite banks routinely collecting data on customer engagement and product usage, organizations often lack quantitative insight into which specific behaviors drive retention, clarity on whether product depth genuinely reduces churn, and evidence on whether high balances alone are sufficient to ensure loyalty. As a result, retention strategies are frequently generic and misaligned with actual customer behavior. This project evaluates retention through the lens of customer behavior and relationship strength, with three core objectives: evaluating the relationship between engagement and churn, measuring the retention impact of product count and product mix, and identifying disengaged yet high-value customers who represent silent, high-exposure churn risk.

3. Data & Methodology

The dataset comprises 10,000 customer records with 14 original fields, including credit score, geography (France, Germany, Spain), gender, age, tenure, account balance, number of products held, credit card ownership, activity status, estimated salary, and a binary churn indicator. No missing values or duplicate customer records were found. All binary fields (HasCrCard, IsActiveMember, Exited) validated as strictly 0/1, and NumOfProducts fell entirely within the expected 1–4 range. A Year field, constant at 2025 across all records, was dropped as it carried no analytical variance. Notably, 3,617 records (36.2%) carry a Balance of exactly zero; these were treated as genuine zero-balance accounts rather than missing data, since imputing them would misrepresent a substantively meaningful customer state. The overall churn rate across the cleaned dataset is 20.37% (2,037 of 10,000 customers), a class imbalance relevant to interpreting later statistical and model-based results.

3.1 Engagement Segmentation Rules

Every customer was classified into exactly one of four engagement profiles, applied in priority order (first match wins):

Segment	Rule
Active Engaged	IsActiveMember = 1 AND NumOfProducts >= 2
Inactive Disengaged	IsActiveMember = 0 AND NumOfProducts = 1
Active Low-Product	IsActiveMember = 1 AND NumOfProducts = 1
Inactive High-Balance	IsActiveMember = 0 AND Balance >= 75th percentile

1,909 customers (19.1%) were inactive, multi-product, but below the balance threshold, and so did not cleanly match any rule; these were folded into “Inactive Disengaged” since inactivity was judged the dominant churn signal. This is an explicit analyst judgment call, stated here for methodological transparency.

4. EDA Findings

4.1 Engagement Profile Findings

Activity status proved to be the single strongest behavioral split found in the dataset. The Active Engaged segment (25.9% of customers) churns at just 9.7%, less than half the overall rate, while Inactive Disengaged customers — the largest single segment at 44.3% of the base — churn at 27.1%, nearly three times higher. The Inactive High-Balance segment (4.2% of customers) is particularly concerning: despite meaningful deposits, this group churns at 24.6%, confirming that balance alone does not buy loyalty.

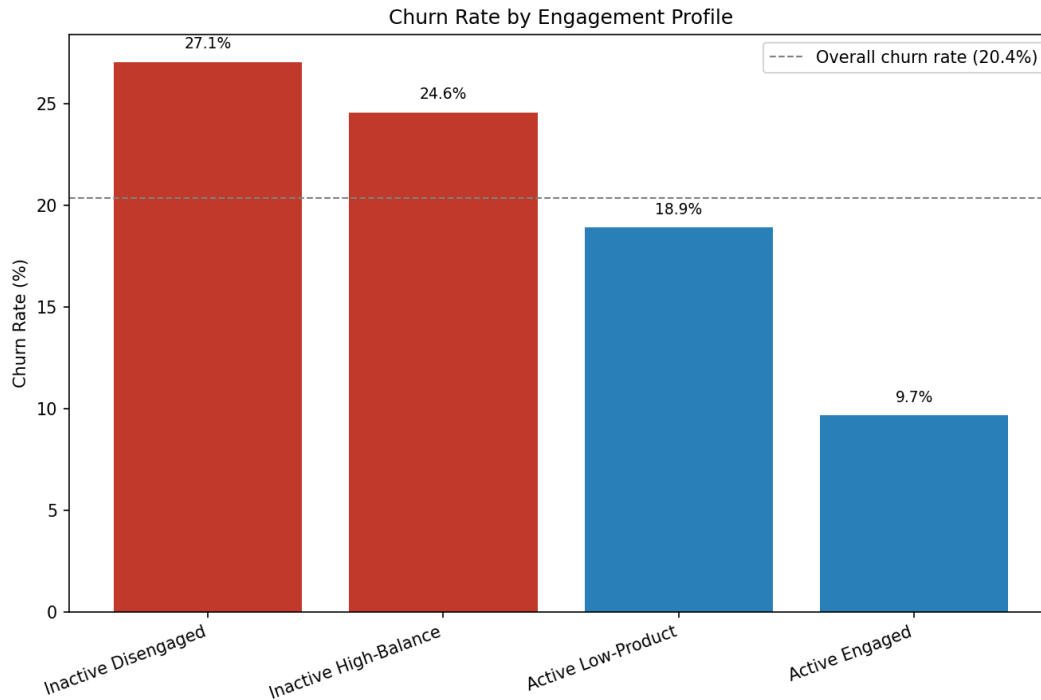


Figure 1. Churn rate by engagement profile, sorted highest to lowest.

Segment	Count	% of Base	Churn Rate
Inactive Disengaged	4,430	44.3%	27.1%
Inactive High-Balance	419	4.2%	24.6%
Active Low-Product	2,563	25.6%	18.9%
Active Engaged	2,588	25.9%	9.7%

4.2 Product Utilization Findings

Product count and churn do not move in a straight line — the relationship is U-shaped. Customers with two products are the stickiest group in the entire dataset (7.6% churn, n=4,590), while single-product customers churn nearly four times higher (27.7%, n=5,084). Beyond two products, however, churn does not continue to fall: it inverts sharply. Three-product customers churn at 82.7% and four-product customers churn at a full 100% (n=60, small group, direction still meaningful). A chi-square test comparing single- versus multi-product customers confirms high significance (chi-square = 342.98, $p < 0.001$), though this aggregate framing masks the more important finding: two products is protective, while three to four products is the strongest churn signal found anywhere in this analysis.

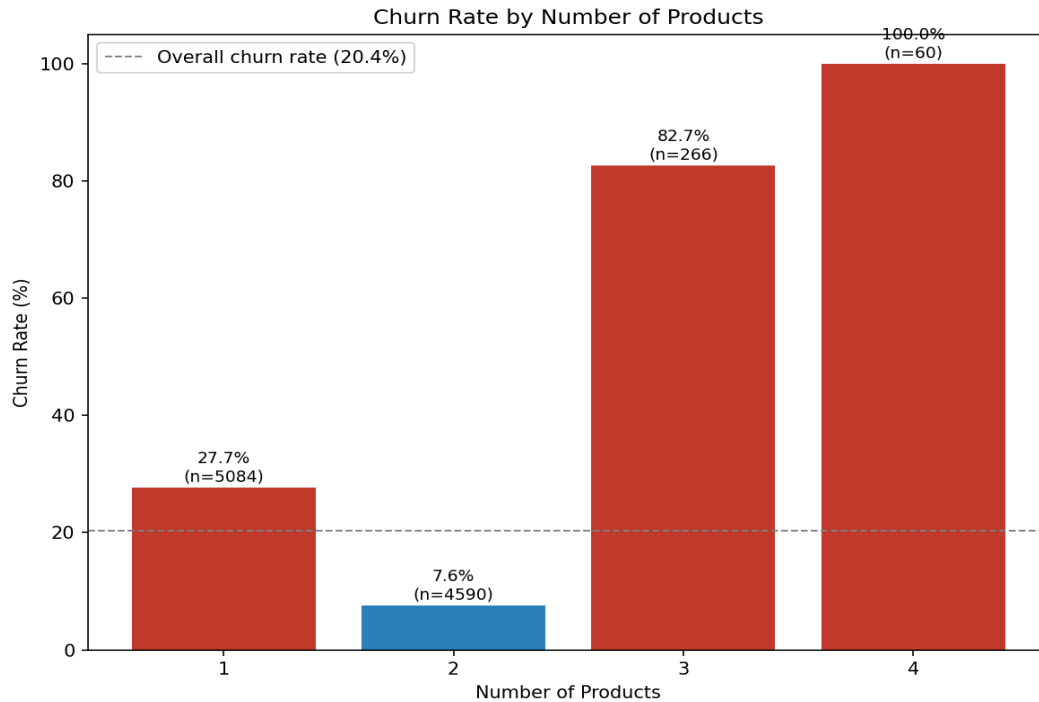


Figure 2. Churn rate by number of products held, with group sizes annotated.

This pattern directly contradicts the standard “more products = more loyalty” assumption past a threshold of two, and it is not a regional or age-group artifact: it holds consistently across France, Germany, and Spain, and across every age band from under-30 to 60+. Two plausible explanations warrant further investigation: (1) customers with three to four products may represent accounts nearing exit that were cross-sold additional products shortly before leaving, making product count a lagging symptom rather than a cause of disengagement; or (2) a servicing or origination artifact specific to three-to-four product bundles (e.g. a discontinued offer or under-supported tier) that correlates with, rather than causes, churn.

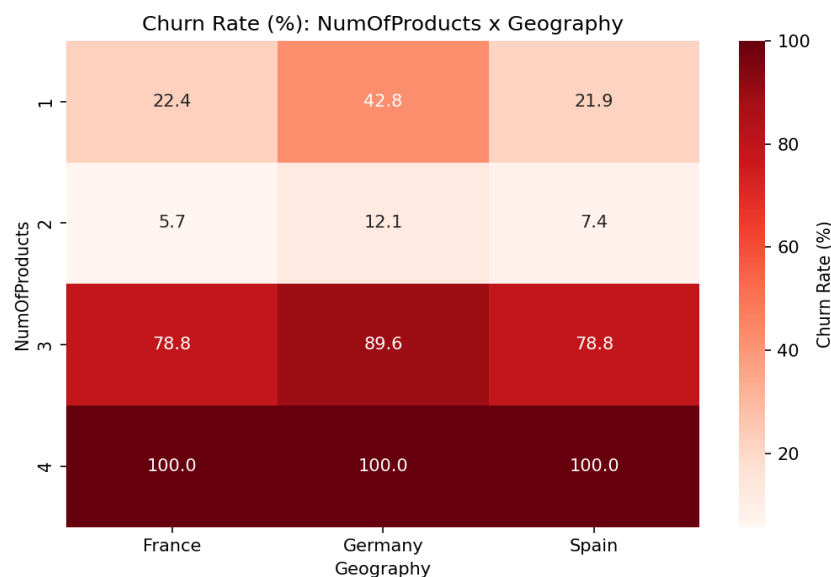


Figure 3. Churn rate (%) by product count and geography — the U-shape holds in every market.

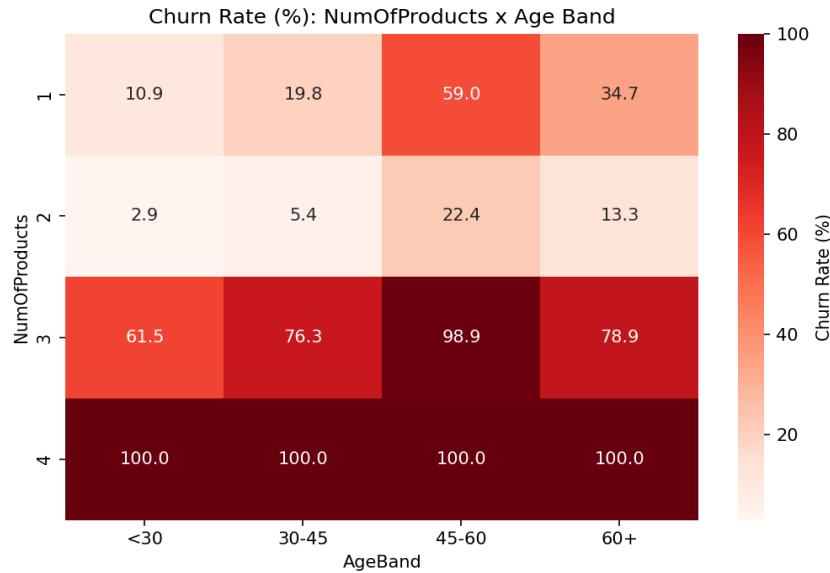


Figure 4. Churn rate (%) by product count and age band — the U-shape holds across every age group.

4.3 Financial Commitment vs. Engagement Findings

At-risk premium customers — defined as customers in the top 25% of account balance who are also disengaged — number 1,247, or 12.5% of the entire customer base. This group holds \$185.6 million in deposits, representing 24.3% of the bank's total balance sheet (\$764.9 million). In plain terms: roughly 1 in 8 customers, holding nearly a quarter of all deposits, are disengaged and at elevated churn risk. This group churns at 30.5%, exceeding the overall rate of 20.4% by more than 10 percentage points.

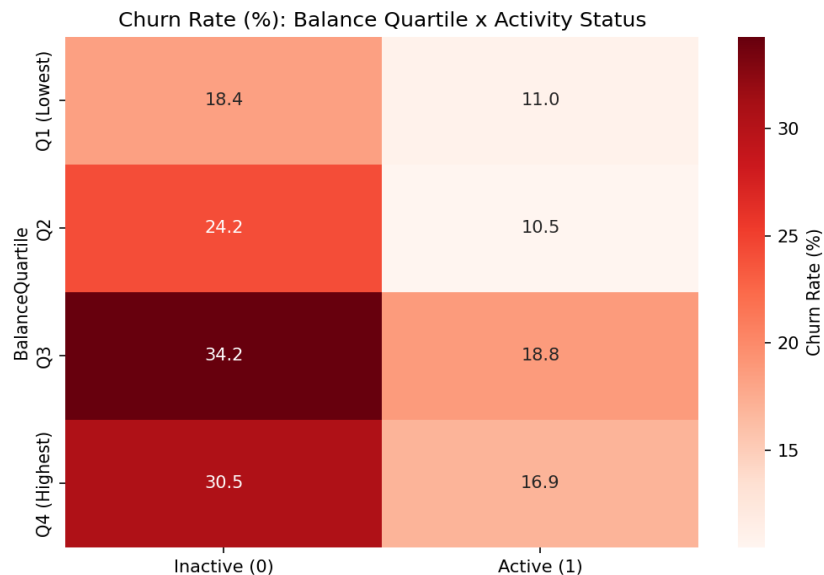


Figure 5. Churn rate (%) by balance quartile and activity status.

The balance-quartile breakdown reinforces this: among inactive customers, churn rises with balance (18.4% in the lowest quartile up to 34.2% in the third quartile), while active customers stay well below the overall average regardless of balance (10.5%–18.8% across all four quartiles). Balance only becomes a

churn risk factor once engagement is already missing — a high-balance, active customer is not at elevated risk, but a high-balance, disengaged customer is the bank's single highest-value flight risk. A supplementary check of balance-to-salary ratio extremity did not yield additional signal beyond this quartile pattern, as the “extreme low” tail was distorted by the large zero-balance population rather than reflecting a genuine outlier group.

5. KPI Framework & Results

Five KPIs were developed to translate the behavioral findings above into reusable, monitorable metrics:

KPI	Formula
Engagement Retention Ratio	churn rate (inactive) / churn rate (active)
Product Depth Index	Pearson correlation of NumOfProducts vs. grouped churn rate
High-Balance Disengagement Rate	% of top-quartile-balance customers who are inactive
Credit Card Stickiness Score	churn rate (no card) minus churn rate (has card)
Relationship Strength Index	$0.4 \times \text{Active} + 0.3 \times (\text{Products}/4) + 0.2 \times (\text{Tenure}/10, \text{ capped}) + 0.1 \times \text{Card}$

5.1 Results by Geography

KPI	Overall	France	Germany	Spain
Engagement Retention Ratio	1.88x	1.84x	1.73x	2.17x
Product Depth Index	+0.858	+0.880	+0.783	+0.889
High-Balance Disengagement Rate	49.9%	50.7%	48.6%	45.0%
Credit Card Stickiness Score	+0.63 pp	+0.32 pp	+0.79 pp	+1.70 pp

5.2 KPI Validation

Four of the five KPIs behaved as expected; one surfaced a genuine exception worth flagging explicitly rather than smoothing over. The Engagement Retention Ratio (1.88x overall) confirms inactive customers churn nearly twice as often as active ones, holding across all three markets. The Credit Card Stickiness Score is positive in all three countries, confirming card ownership is mildly protective everywhere, strongest in Spain (+1.70 pp). The Relationship Strength Index is the clearest success: churn falls monotonically across every quartile, from 28.8% in the weakest-relationship quartile down to 12.7% in the strongest — a clean validation that the composite weighting captures real retention signal.

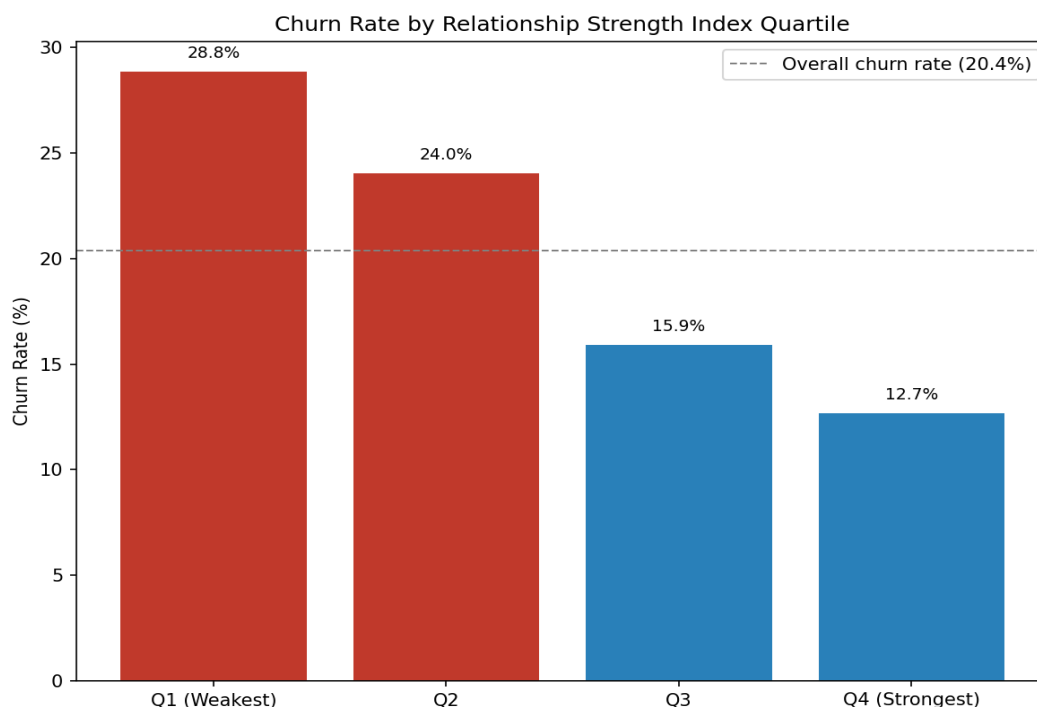


Figure 6. Churn rate by Relationship Strength Index quartile — a clean, monotonic decline.

The Product Depth Index is the one KPI that inverts naive expectation, by design and not by error: it was defined as the correlation between product count and grouped churn rate, which directly encodes the Section 4.2 finding. The correlation is strongly positive (+0.858 overall, positive in all three countries), meaning more products associates with higher churn across the dataset as a whole, because the index is dominated by the extreme three-to-four product cliff. This KPI should therefore be read as “product-count risk concentration,” not “product depth benefit” — a distinction worth stating explicitly so the metric is not misread at a glance. Finally, the High-Balance Disengagement Rate (49.9% overall, roughly a coin flip, consistent 45–51% across markets) reinforces that balance alone says nothing about engagement; it must be paired with activity status to identify real risk.

6. Predictive Validation

As an independent check on the KPI framework, a Random Forest classifier (200 trees, max depth 8, class-balanced weighting, 80/20 train-test split) was trained to predict churn from all available customer attributes. This was treated as supporting evidence, not a deployment-ready model.

Metric	Value
Accuracy	81.95%
Precision (Exited = 1)	54.46%
Recall (Exited = 1)	69.04%

Given the approximately 20% churn class imbalance, accuracy alone is not the meaningful number here — a model that always predicted “stayed” would already score roughly 80%. Precision and recall on the

churned class are more informative: the model correctly flags roughly 7 in 10 actual churners, at the cost of a moderate false-positive rate, a reasonable trade-off for retention-targeting use cases where missing a churner is costlier than an unnecessary retention outreach.

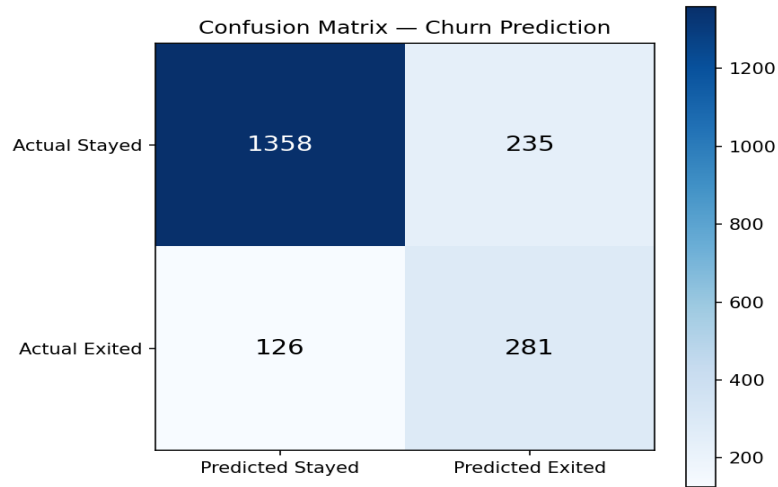


Figure 7. Confusion matrix on the held-out test set ($n=2,000$).

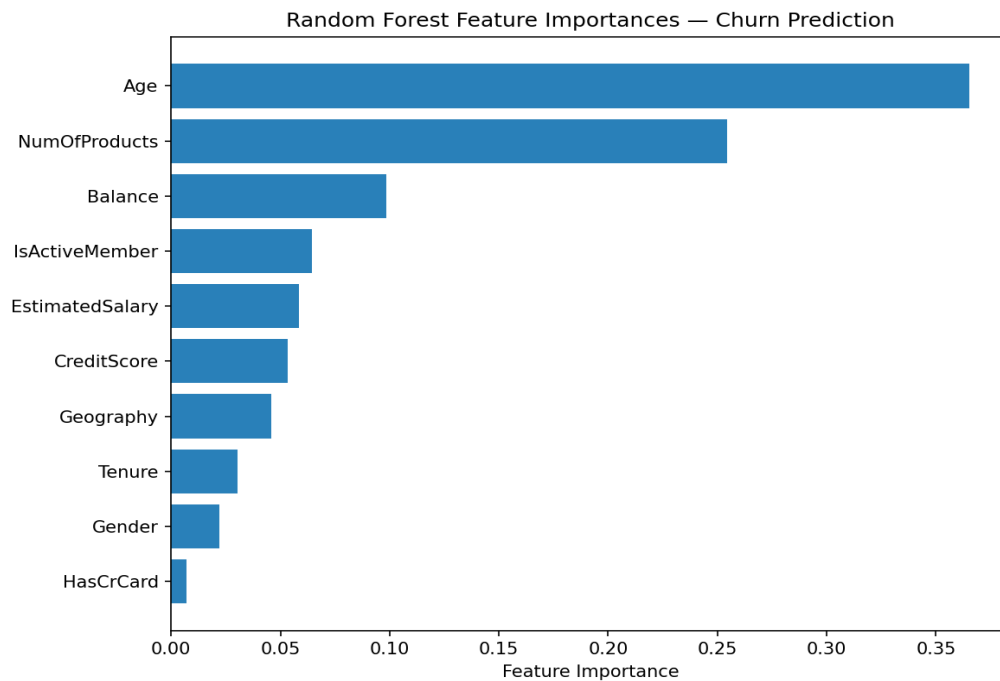


Figure 8. Random Forest feature importances, ranked highest to lowest.

The feature importances partially support and partially complicate the KPI framework. NumOfProducts ranks as the second most important feature overall (importance 0.255), directly corroborating the product-count cliff identified in Sections 4.2 and 5.2 as a dominant, data-confirmed churn driver. IsActiveMember also registers as meaningfully important (ranked 4th), supporting the engagement-based KPIs. However, Age emerges as the single strongest predictor by a wide margin

(importance 0.365, ahead of every engagement or product variable) — a feature not incorporated into any of the five KPIs. This is a genuine gap worth flagging: age's dominance suggests older customers may be churning for reasons the current engagement/product framework does not capture, such as retirement, life-stage banking needs, or a support experience that skews poorly for older customers. Notably, EstimatedSalary ranks low (5th), reinforcing the core project thesis that raw salary is not a meaningful churn driver on its own, in contrast to behavioral and structural variables like age and product count.

7. Recommendations

- **Prioritize re-engagement over pure cross-sell.** Since activity status alone moves churn risk further than adding a second or third product (9.7% vs. 27.1% churn between active and inactive segments), retention budget should favor activation nudges and usage incentives for the Inactive Disengaged segment (44.3% of the base) ahead of generic cross-sell campaigns.
- **Cap cross-sell at two products and investigate the three-to-four product cliff as a priority, not a footnote.** Given churn jumps from 7.6% at two products to 82.7–100% at three to four products, sales teams should stop treating additional products past two as an automatic loyalty win, and a root-cause investigation (servicing quality, bundling practices, or exit-driven cross-sell timing) should be commissioned immediately.
- **Launch a dedicated retention program for the 1,247 at-risk premium customers.** This group holds \$185.6 million (24.3% of deposits) while disengaged and churning at 30.5%. Given the concentrated financial exposure, this segment justifies white-glove relationship management rather than mass-market retention tactics.
- **Adopt the Relationship Strength Index as an ongoing monitoring metric,** since it cleanly and monotonically predicts churn (28.8% to 12.7% across quartiles), and route the bottom quartile into proactive outreach workflows.
- **Investigate age as a retention factor** in a follow-up study, given it was the strongest predictor in the Random Forest model yet sits entirely outside the current KPI framework — an age-banded engagement KPI is a natural next iteration.

8. Limitations

- **Single-snapshot dataset.** The data represents one point in time (the Year field was constant), so no trend, cohort, or time-to-churn analysis was possible; findings describe association, not confirmed causal direction.
- **Analyst-defined thresholds.** The EngagementProfile segmentation rules and the balance quartile cutoffs used throughout are reasoned judgment calls, not universal definitions; 19.1% of customers required a fallback classification rule.
- **No causal mechanism confirmed for the product-count cliff.** The three-to-four product churn spike is a robust statistical pattern, but this analysis cannot distinguish whether products caused disengagement, were a symptom of it, or reflect an unrelated servicing issue.
- **Predictive model is illustrative, not production-ready.** The Random Forest model was built for KPI validation, not deployment; it was not tuned, cross-validated across multiple splits, or tested for fairness across demographic groups.

9. Conclusion

This project reframes customer churn from a behavioral and relationship-strength perspective rather than a demographic one. The evidence is consistent across every analytical lens used — segmentation, cross-tabulation, KPI construction, and independent model validation — that engagement status and product-count structure predict churn far more reliably than balance or salary alone. The clearest actionable finding is that disengagement, not low wealth, is the bank's core retention problem: the highest-exposure customers are not the poorest, but the wealthiest ones who have stopped engaging. Equally important is the discovery that product depth is not uniformly protective; retention strategy must treat the two-product tier as the target state rather than assuming that more is always better. Together, these findings support a shift from generic, demographic-led retention campaigns toward engagement-driven, exposure-weighted interventions — with age flagged as a priority area for future research beyond the scope of the current KPI framework.